

Syvis

A visual code editor for Lively

Software Architecture Group

Prof. Dr. Robert Hirschfeld

Web Based Development Environments

2017-2018

Demo

1. Add.js - Same function, several ways to format it
2. Gotchas.js - Hard to spot bugs
3. Primitives.js - Responsive layout and emojis
4. Loops.js - Custom syntax style and pretty objects
5. Gotchas.js - Fix the bugs
6. Large.js - Demonstrate good performance

/examples/gotchas.js visualize

```
1 function getDescriptionBroken (name, features) {  
2   return  
3   `This is the description of product ${name}.  
4   It's features are: ${features.join()}`  
5 }  
6  
7  
8 for (let item of [1, 2, 3]);  
9   console.info(item);  
10  
11  
12 const math = 3 + 4 * 5 + 6 * 7 * 8  
13
```

/examples/gotchas.js edit

getDescriptionBroken (name, features)

function

This is the description of product \${ name }. It's features are: \${ features
.join() }

for item of
1
2
3
<empty statement>

console
.info(item)

math := ((3 + (4 × 5)) + ((6 × 7) × 8))

The Problem

1. **Plain text can be hard to comprehend**

=> Formatting & syntax highlighting try to mitigate this problem

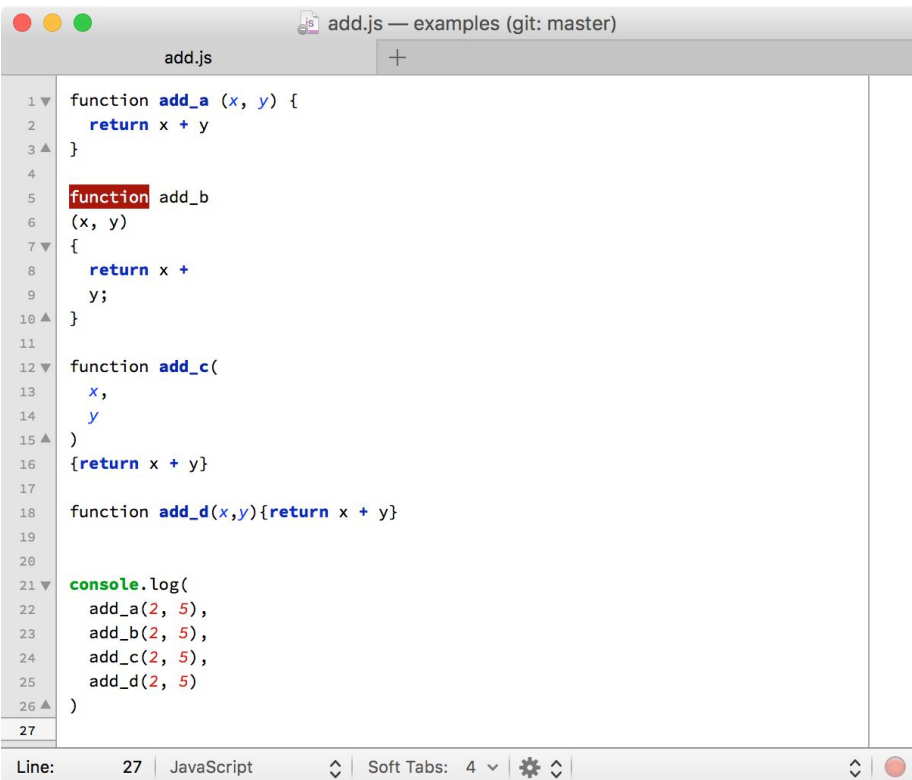
2. **Formatting is decided by the author and not the reader**

=> Code is less comprehensible in unfamiliar formatting which leads to decreased development speed and higher error rates

3. **Semantically equivalent code can be written in vastly different textual representations**

=> Makes it hard to reason about code

Instead: Syntax Visualization



```
1 function add_a (x, y) {  
2   return x + y  
3 }  
4  
5 function add_b  
6 (x, y)  
7 {  
8   return x +  
9   y;  
10 }  
11  
12 function add_c(  
13   x,  
14   y  
15 )  
16 {return x + y}  
17  
18 function add_d(x,y){return x + y}  
19  
20  
21 console.log(  
22   add_a(2, 5),  
23   add_b(2, 5),  
24   add_c(2, 5),  
25   add_d(2, 5)  
26 )  
27
```

Line: 27 | JavaScript | Soft Tabs: 4



Architecture

Syvis

let value = 6 * 7

let value = 42

Initial
Code

New
Code

Serialize

Transform to
AST

['div', {id: '... <div id="...

Transform to
virtual DOM

Render in
document

```
"type": "VariableDeclarator",  
"id": {  
  "type": "Identifier",  
  "name": "value"  
},  
},
```

Patch the AST
(original formatting is
lost)

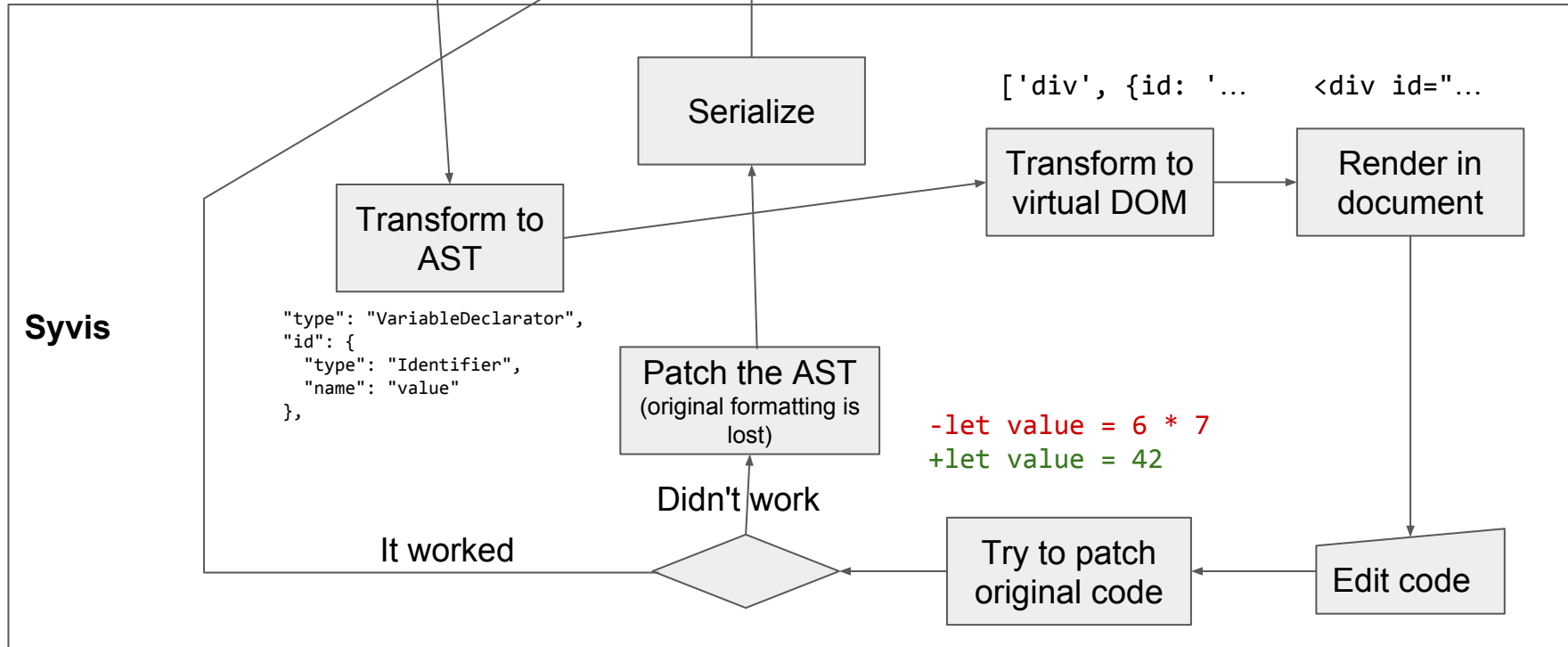
-let value = 6 * 7
+let value = 42

Didn't work

It worked

Try to patch
original code

Edit code



Challenges

- **Rendering is complex**
 - ~120 commits just to get it working
 - 64 node visualizers and still counting
 - (Almost) unlimited edge cases
- **Developer needs to get used to it**
- **Tooling needs to be built**

Future Work

1. **Better editing**
2. **Several Themes + Selector**
3. **Different input elements for different data types**
 - a. **Slider**
 - b. **Calendar widget**
4. Drag and drop reordering
5. Multi language support with integrated cross compilation
6. AST all the way
7. Multi language rendering (Math formulas, Latex, Markdown)

Insert Integer Widget

Insert Table Widget

Insert Graph Widget

1 const aNumber =

-

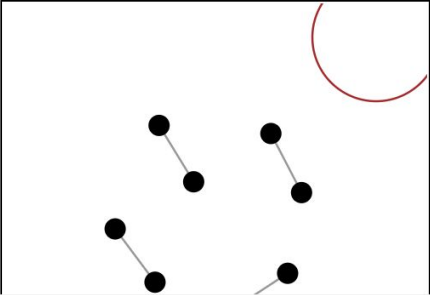
0

+

2

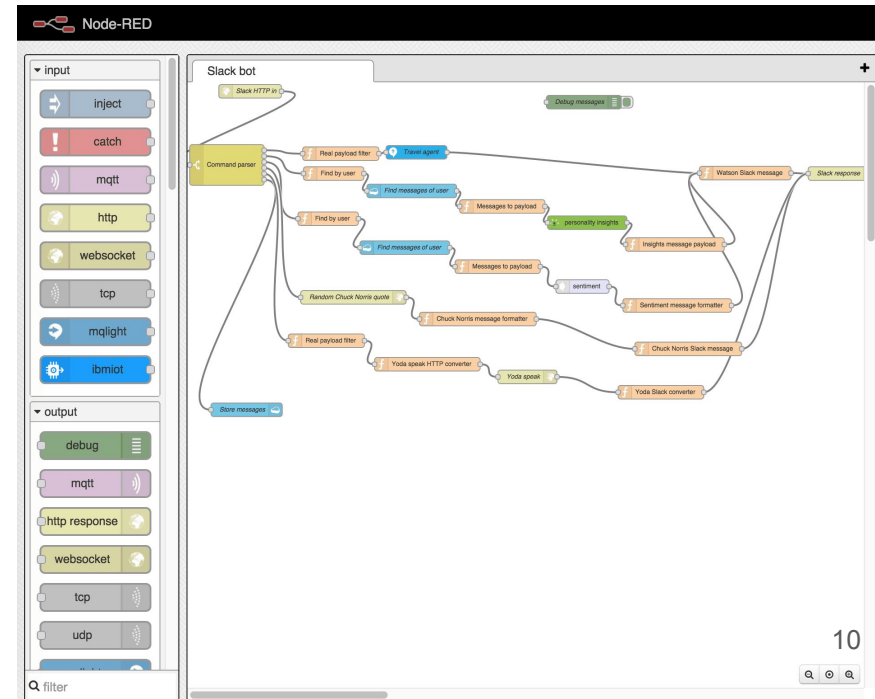
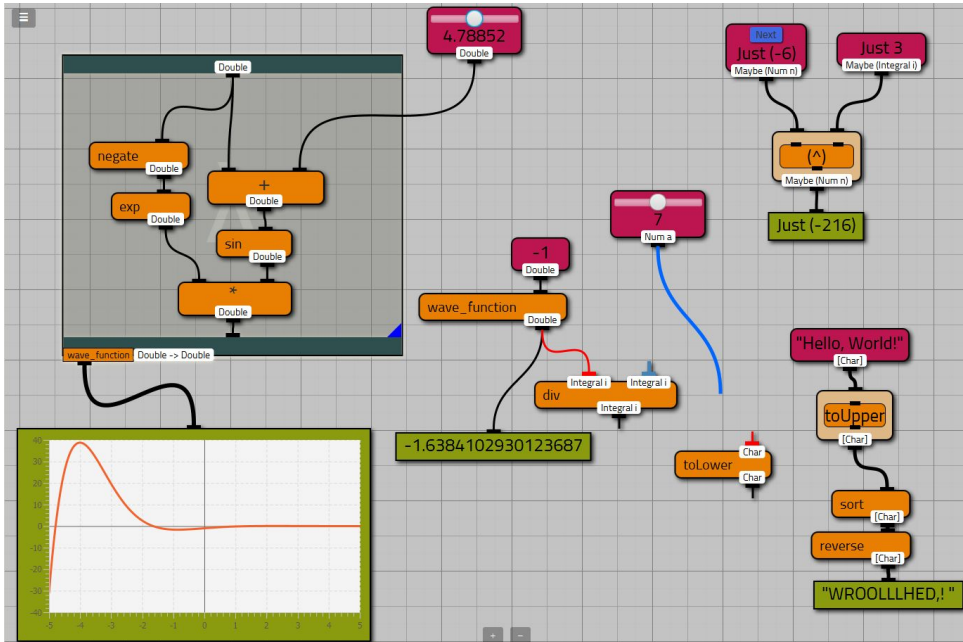
3 const aTable =

Na	Age	Size
Te	45	122
Te	45	122
Te	45	122



Probably Not Ideal: Flow Graphs

Still hard to read
Still unlimited different representations



Inline Widgets



attempt at CodeMirror + in-place MathJax [\[source on github\]](#)

MathJax version: 2.7.1 | CodeMirror version: 5.26.0 Load (for any other versions edit URL params)

Finished rendering math in 3148ms.

```

1 # Inline math
2  $a < b > c$ 
3 > xXxXxXxXxXxXxXxXxXxXxXxXxX
4 Normal-size line with  $n = 2$  compact formulas:  $x * y$ . There was a  $\sim^*$  in that formula but this text should not be italic.
5 xXxXxXxXxXxXxXxXxXxXxXxXxX
6 >> Foo  $\sum_i \frac{1}{i}$  bar.
7 xXxXxXxXxXxXxXxXxXxXxXxXxX
8
9 ## Misc MathJax extensions
10
11 
$$\begin{array}{ccc} A & \longleftrightarrow & B \longrightarrow C \\ \text{AMScd:} & \parallel & \uparrow \\ & D \equiv & E \end{array}$$

12
13 cancel:  $\int \frac{\cos \frac{A}{X}}{X^C}$ 
14
15 mhchem:  $A \xrightarrow{+H_2O} B$ 
16
17 *et cetera...* (No thanks to this library, I simply enabled them in \[MathJax config\]\(http://docs.mathjax.org/en/latest/tex.html\).)
18
19 ## Display math
20
21 
$$\sum_c \frac{1}{a}$$

22
23 
$$\nabla \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} = \frac{4\pi}{c} \vec{j}$$

24
25 
$$\nabla \cdot \vec{E} = 4\pi \rho$$

26
27 
$$\nabla \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = \vec{0}$$

28
29 
$$\nabla \cdot \vec{B} = 0$$


```

```

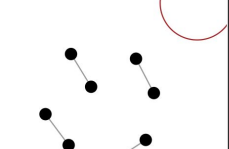
1 var widgets = [ ]
2 function updateHints() {
3     editor.operation(function(){
4         for (var i = 0; i < widgets.length; ++i)
5             editor.removeLineWidget(widgets[i]);
6         widgets.length = 0;
7
8         JSHINT(editor.getValue());
9         for (var i = 0; i < JSHINT.errors.length; ++i) {
10             // ...
11         }
12     });
13 }

```

Number =

Table =

Nat	Age	Size
Test	45	122
Test	45	122
Test	45	122



```

r = <<0>>
= <<matrix(QQ,[[ "Name", "Age", "Size"],[ "Test",45,122],[ "Test",45,122]])>>
= <<{"widget": "graph", "vertices":8, "edges":[[1,0],[5,4],[6,3],[7,2]]}>>

```

Edit using text...

```
; This is a comment. It just stays where it is.

; We can have literals of various types
42      ; number
"hello"  ; string
#\m      ; character
#t       ; boolean
quuz     ; symbol

; we can have if expressions
(if (positive? -5) (error "doesn't get here") 2)

; we can have cond expressions
(cond
  [(positive? -5) (error "doesn't get here")]
  [(zero? -5) (error "doesn't get here, either")]
  [(positive? 5) #t])

; we can have lambda expressions
(lambda (x y) (+ x y))

; we can define a variable or two
(define FIRST-NAME "John")
(define LAST-NAME "Doe")

; we can have structures
(define-struct person (first-name last-name age country))

; which we can then make instances of
(define john (make-person FIRST-NAME LAST-NAME 28 "USA"))
```

or edit using blocks!

```
; This is a comment. It just stays where it is.

; We can have literals of various types
42      ; number
"hello"  ; string
#\m      ; character
#t       ; boolean
quuz     ; symbol

; we can have if expressions
```



```
; we can have cond expressions
```



☒ Block Mode

Use AST (not the Code)

Parser produces the (beautiful) syntax tree

```
1 const value = x + y
```

No error

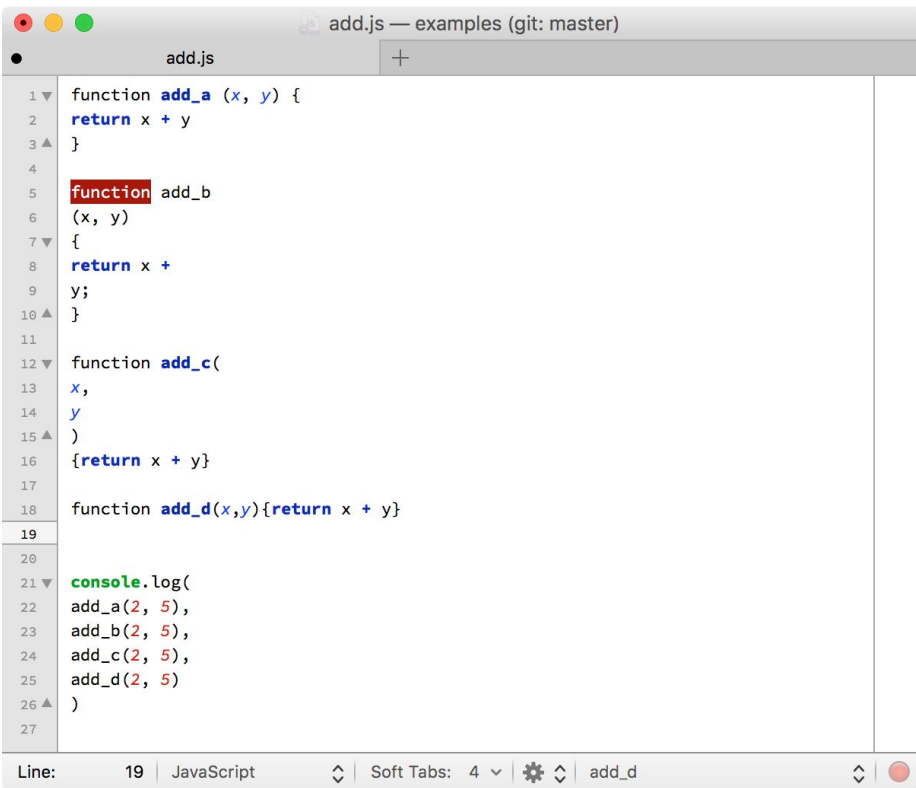
Syntax node location info (start, end):

- ☐ Index-based range
- ☐ Line and column-based
- ☐ Attach comments

Syntax Tree Tokens

```
{
  "type": "Program",
  "body": [
    {
      "type": "VariableDeclaration",
      "declarations": [
        {
          "type": "VariableDeclarator",
          "id": {
            "type": "Identifier",
            "name": "value"
          },
          "init": {
            "type": "BinaryExpression",
            "operator": "+",
            "left": {
              "type": "Identifier",
              "name": "x"
            },
            "right": {
              "type": "Identifier",
              "name": "y"
            }
          }
        }
      ]
    }
  ],
  "kind": "const"
},
"sourceType": "script"
}
```

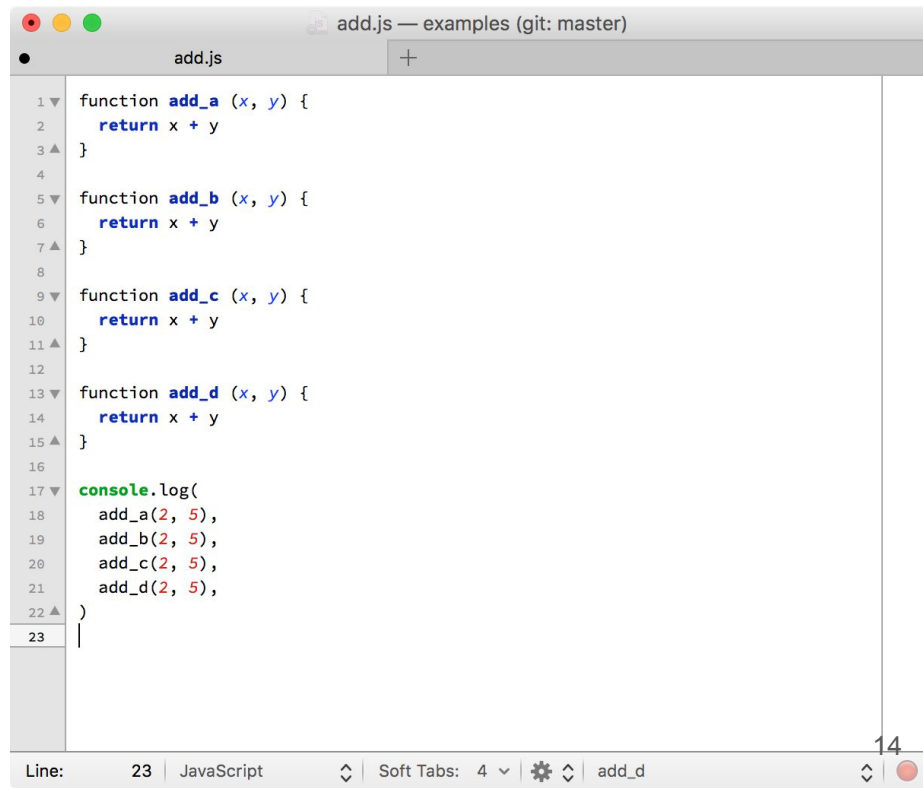
Formatting Should be a User Setting



The screenshot shows a code editor window titled "add.js — examples (git: master)". The code is as follows:

```
1 function add_a (x, y) {  
2   return x + y  
3 }  
4  
5 function add_b  
6 (x, y)  
7 {  
8   return x +  
9   y;  
10 }  
11  
12 function add_c(  
13 x,  
14 y  
15 )  
16 {return x + y}  
17  
18 function add_d(x,y){return x + y}  
19  
20  
21 console.log(  
22 add_a(2, 5),  
23 add_b(2, 5),  
24 add_c(2, 5),  
25 add_d(2, 5)  
26 )  
27
```

The status bar at the bottom indicates "Line: 19 | JavaScript | Soft Tabs: 4 | add_d".



The screenshot shows the same code editor window with the same code, but with consistent formatting:

```
1 function add_a (x, y) {  
2   return x + y  
3 }  
4  
5 function add_b (x, y) {  
6   return x + y  
7 }  
8  
9 function add_c (x, y) {  
10  return x + y  
11 }  
12  
13 function add_d (x, y) {  
14  return x + y  
15 }  
16  
17 console.log(  
18   add_a(2, 5),  
19   add_b(2, 5),  
20   add_c(2, 5),  
21   add_d(2, 5),  
22 )  
23 |
```

The status bar at the bottom indicates "Line: 23 | JavaScript | Soft Tabs: 4 | add_d".